

ZOO GARDEN CONSOLE APPLICATION

OOP PROJECT - 1

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INTRODUCTION

The purpose of this project is to design a Zoo Garden console application using Object-Oriented Programming principles in Python. The application will allow zoo staff to manage animals and zoo information through a console menu.

THIS TASK IS BASED ON PYTHON (OOP), GIT AND GITHUB



EPICS

Epic 1: Animal Management

- Add animal
- View animal
- Update animal
- Remove animal

Epic 2: Employee Management

- Add employee
- View employee
- Remove employee

Epic 3: Zoo Information

- View zoo statistics
- Search for animals

4. User Stories

US1 / As a zookeeper, I want to add a new animal so that the zoo records remain updated.

US2 / As a zookeeper, I want to update animal information so that records stay accurate.

US3 / As a visitor, I want to view animal information so that I can learn about the animals.

US4 / As a manager, I want to remove animal records so that outdated information is deleted.

US5 / As a manager, I want to view zoo statistics so that I can monitor zoo operations.

add a new animal

update animal information



zookeeper

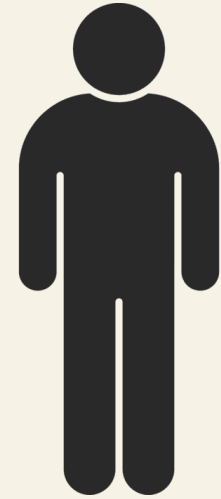
view animal information



visitor

remove animal records

view zoo statistics



Manager

MOCKUP- OVERVIEW

System main view

```
=====
      ZOO GARDEN
=====
1. Add Animal
2. View Animals
3. Update Animal
4. Remove Animal
5. View Zoo Statistics
6. Exit
Choose an option:
```

STATISTICS

```
===== ZOO STATISTICS =====
Total Animals: 2
```

Remove elements

```
Choose an option: 4

Animal #1
Name: mike
Species: lion
Age: 15

Animal #2
Name: king
Species: lion
Age: 9
Enter animal number to remove: 1
mike removed successfully!
```

View data - Reading

```
Choose an option: 2

Animal #1
Name: mike
Species: dog
Age: 3

Animal #2
Name: king
Species: lion
Age: 9
```

Update

```
Choose an option: 3

Animal #1
Name: mike
Species: dog
Age: 3

Animal #2
Name: king
Species: lion
Age: 9
Enter animal number to update: 1
New name: mike
New species: lion
New age: 15
Animal updated successfully!
```

Add

```
Choose an option: 1
Enter animal name: mike
Enter species: dog
Enter age: 3
Animal added successfully!
```

CONCLUSION

The Zoo Garden console application was designed using Object-Oriented Programming principles in Python. During the planning phase, epics, user stories, use case diagrams, and console menu mock-ups were created to define the system requirements and functionality.

The application allows users to manage animal records through features such as adding, viewing, updating, and removing animals, as well as viewing basic zoo statistics. The use of classes and objects helped organize the program structure and model real-world entities within the zoo.

This project demonstrates the application of OOP concepts and provides a simple console-based system for zoo management.

THANK YOU ALL